

1. Differentiate each of the following. Simplify your answers.

$$f(x) = \frac{3x}{5} - \frac{5}{3x} - 4 + 2\sqrt[4]{x}$$

$$y = \frac{20x^5 + 8x^2 + 4x}{6x^3}$$

$$y = 4x^{\frac{2}{3}}(x - 5\pi^3)$$

$$f(x) = \frac{\sqrt{3}x^4}{3} + \frac{7}{3x^{\frac{2}{3}}} + 21\sqrt[17]{x} - 3\pi^4$$

$$y = 2x\sqrt{x}(x + 7\sqrt{x} - 8)$$

$$g(x) = \frac{3\pi^{\frac{2}{3}}(4x^2 + 9\sqrt{x})}{x}$$

2. Determine the equation of the tangent line to the curve $y = x + \frac{1}{x^2}$ that is parallel to the line $x + y + 3 = 0$.